

Roha Patel

Nashville, TN | +1 (570) 983-6440 | rohapatel64@gmail.com | LinkedIn: [roha-patel](#) | Github: [Roha-Patel](#)

Education

Data Analytics and Visualization (Bootcamp) | University of California, Berkeley, CA 2024
Bachelor of Science, Physics | Sardar Patel University, India May 2021

Certifications

Quality Assurance (QA) & Selenium 2025
Web Development | Sardar Patel University, India May 2021

Core Competencies

Programming Languages: SQL, Java, Python, Ruby, C
Automation Tools: Selenium WebDriver, Jenkins, Eclipse, IntelliJ IDEA, RubyMine
Databases: OracleDB, MSSQL Server, MySQL, PostgreSQL, MongoDB, SQLAlchemy
CI/CD Tools & Frameworks: Git, GitHub, GitLab, Jira, Postman, Jenkins, SwaggerUI

Projects

Selenium Automation Testing – Banking Application (GURU99) Oct 2025

- Gained **hands-on experience in Selenium automation** through a real-time project simulating a corporate testing environment.
- Developed and executed **Selenium WebDriver test scripts** using **TestNG** for test management, verification, and reporting.
- Implemented **data-driven testing** by parameterizing scripts with **Excel**, adding validation steps and screenshot capture for results tracking.
- Enhanced automation proficiency by comparing scripts with **expert-level solutions**, aligning with **industry-standard QA practices**

LA Crime Statistics | [Github](#) | [Python, Flask, PostgreSQL, Pandas, SQLAlchemy] May 2024

- Developed a **Flask-based web app** to visualize Los Angeles crime data using interactive maps and charts.
- Cleaned and loaded **LAPD Open Data** into **PostgreSQL** with **Pandas** and **SQLAlchemy**, ensuring accurate and structured datasets.
- Built **interactive visualizations** using **Leaflet**, **Plotly.js**, and **Chart.js**, enabling users to explore crime trends by date, area, and type.
- Implemented **data filtering, pagination, and time-based analysis features**, improving usability and data exploration efficiency.

Crowdfunding ETL | [Github](#) | [Python, Pandas, SQLAlchemy, PostgreSQL] April 2024

- Built an **ETL pipeline** in Python and Pandas to extract and clean crowdfunding data from multiple Excel sources.
- Created **four structured CSV files** using **dictionary methods and regex**, ensuring clean and consistent data.
- Designed an **ERD and database schema** in QuickDBD, defining clear primary and foreign key relationships.
- Imported CSV files into **PostgreSQL**, verified data accuracy, and delivered a fully functional **Crowdfunding database**.

Global Energy Consumption | [Github](#) | [Python, Pandas, Matplotlib, SciPy, NumPy, Scikit-learn] March 2024

- Analyzed global energy and CO₂ data using **Python** and **Pandas** to find key correlations.
- Identified a strong link between **CO₂ emissions** and **energy consumption**, with electricity as the major source.
- Merged and cleaned **multinational datasets** in **Jupyter Notebook** for trend analysis.
- Derived insights on **sustainable energy**, highlighting low-emission alternatives and global shifts.

Work Experience

Scientist | PACE Analytics, Mount Juliet, TN Nov 2024 - Present

- Maintained **high accuracy and attention to detail** while performing soil and water quality tests.
- Documented results following **standard procedures**, ensuring reliability and compliance.
- Collaborated with teams to **analyze test outcomes** and resolve discrepancies effectively.
- Managed multiple projects, demonstrating **strong time management, organization, and adaptability**.